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Introduction:

Convert multiple strain measurements at arbitrary angles to principal strain and angle of principal strain and determine stress.

The program 'converge.exe' has been designed to assist in the interpretation of closure station data. For input, the program takes three values of strain, a direction angle for each strain, Young's Modulus and Poisson's Ratio. A default value for Poisson's Ratio is set at .2, however, the user should realize that this value can have a significant effect on results.

Program output consists of principal strain values and orientation, as well as principal stresses. The theory used for obtaining principal strains and orientation is based on a matrix inversion and solution taken from Introduction to Rock Mechanics by Goodman (1989). The resolution of principal stresses from strains applies the Kirsch solution taken from Rock Mechanics for Underground Mining by Brady & Brown, (1992). This solution assumes a circular tunnel geometry and an elastic rock mass response. When the elastic response of the rock mass appears to have been exceeded, the orientation and magnitude of principal strains may still provide useful data.

This program has been developed by Milne & Milne Engineering in conjunction with the UBC Mining-Geotechnical Group. Assistance has been provided by the

Noranda Technology Centre and Energy Mines and Resources.

As of June 2010 work was started to convert the core program to a version 2.x The central algorithms that do the mathematics remain the same only a windows forms GUI was built and some file handling to read different popular file formats (.CSV, .XLS and XLSX specifically).

If you find this

program of use, please send 12 chocolate chip cookies (no nuts please) to

Milne & Milne Engineering, 11067 129 Street, Surrey BC V3T 3J3

System Requirements:

-A Pentium class computer with 1gb or better RAM capacity.

-Windows OS (Windows XP Windows Vista or Windows 7) Will not run correctly

Under WINE on linux etc.

-The input data files need to be .CSV .XLS or .XSLX so though not required EXCEL or OpenOffice spread sheet would be useful for checking file format.

-This is managed code so it must run under .NET framework 2.0 or better (if you are up to date on your Microsoft updates you should be good)

Installation:

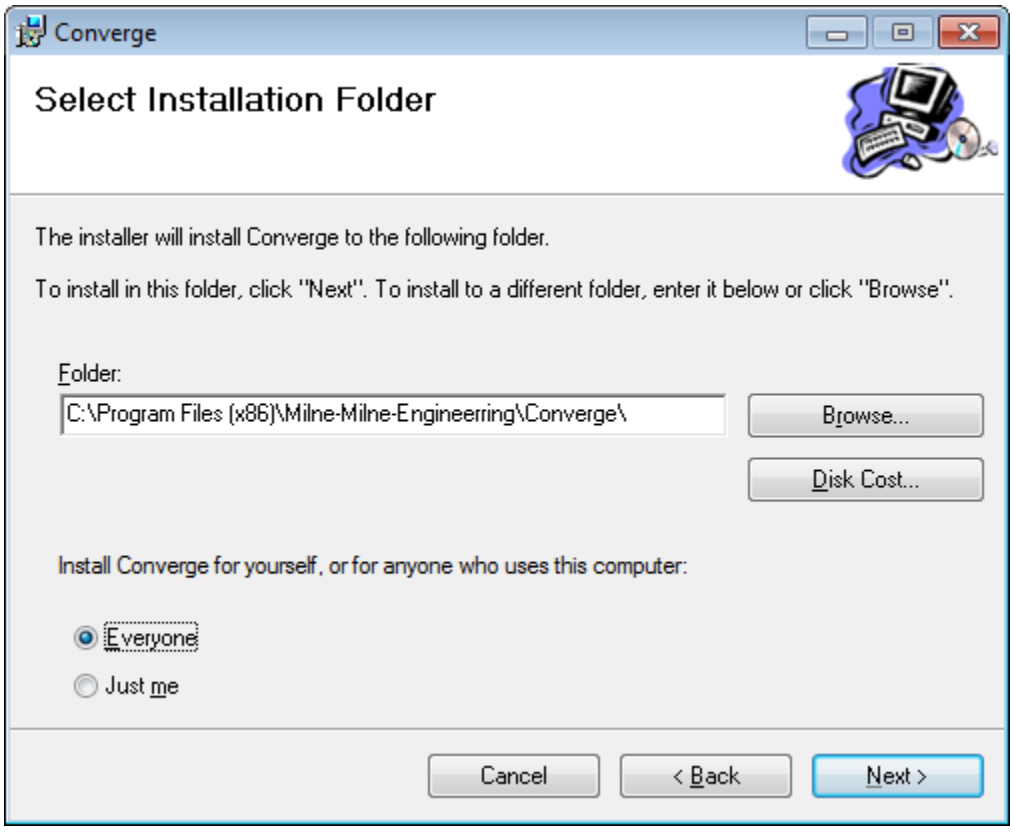
Files provided are :

ConvergeSetup.msi

Setup.exe

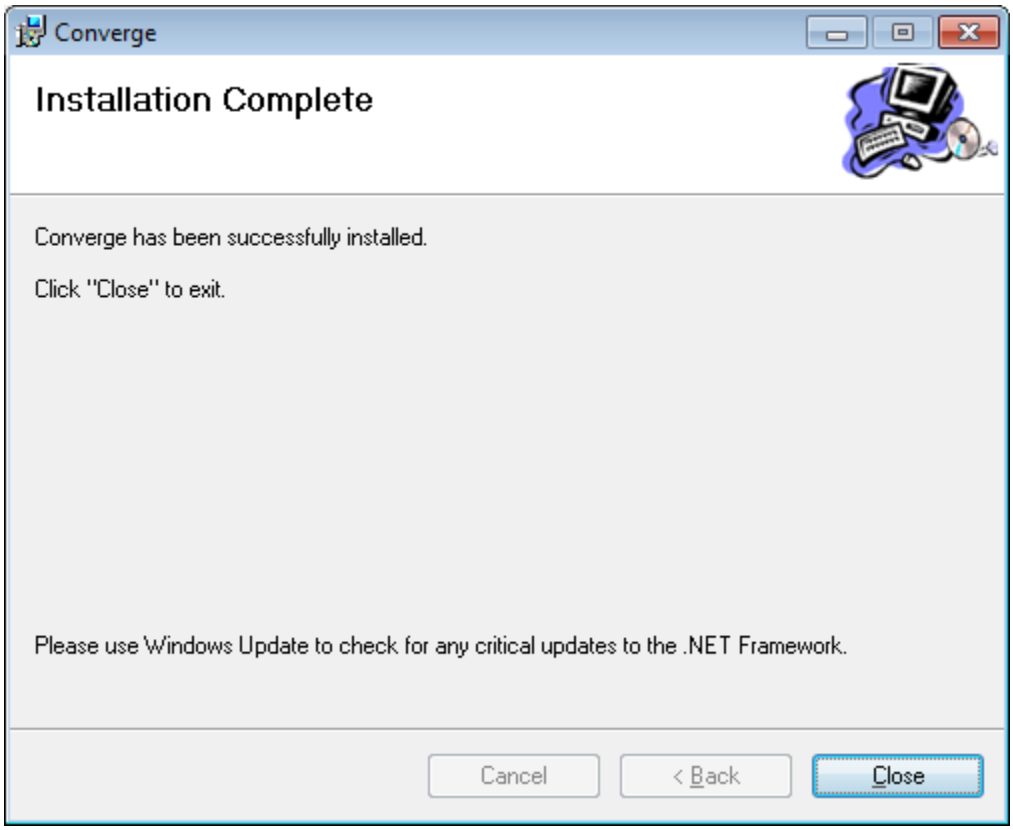
ConvergeDocumentation.PDF (this document)

The installation should be quite simple double click the MSI file and you should see:



Select the folder you want (Not above it says Program Files (x86) on a 64 bit machine location is immaterial)

Follow the prompts along and you should see:



As well you will find Desktop Icons and start bar icons looking like this



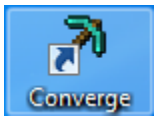
As well once installed some sample data files are provided in the directory like:

C:\Program Files (x86)\Milne-Milne-Engineerring\Converge\Sample Files

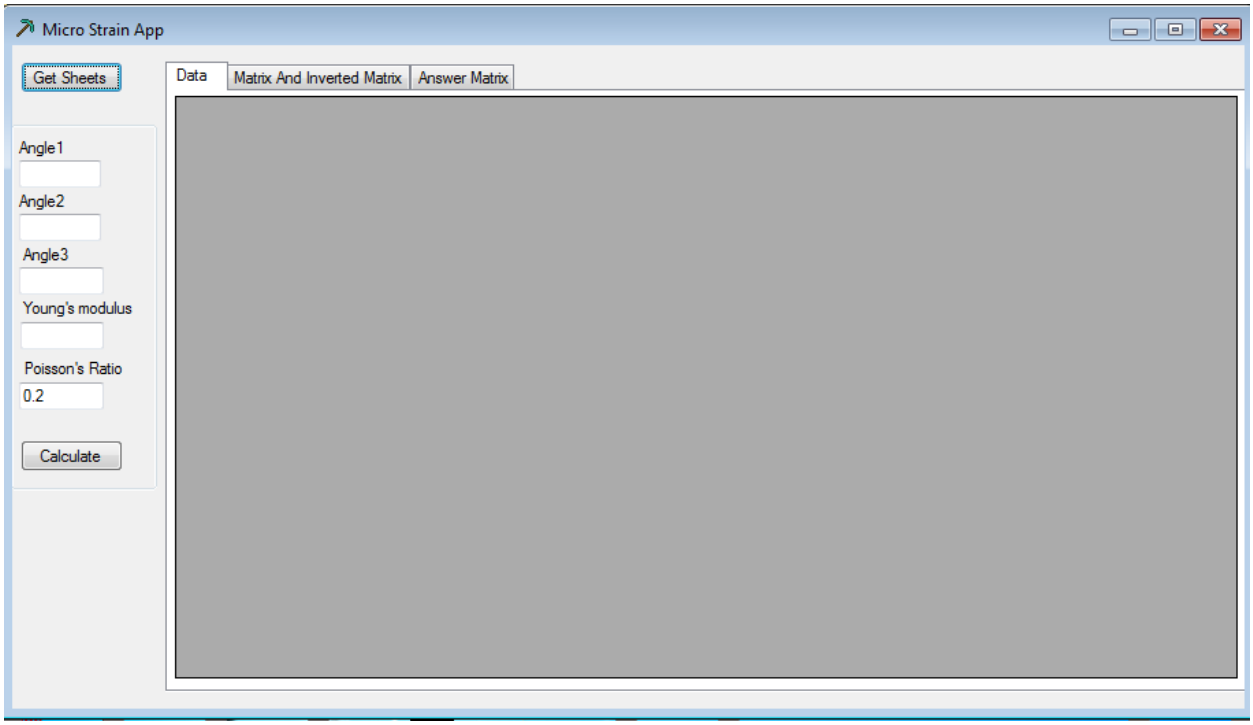
Path will vary depending on your choices.

USAGE:

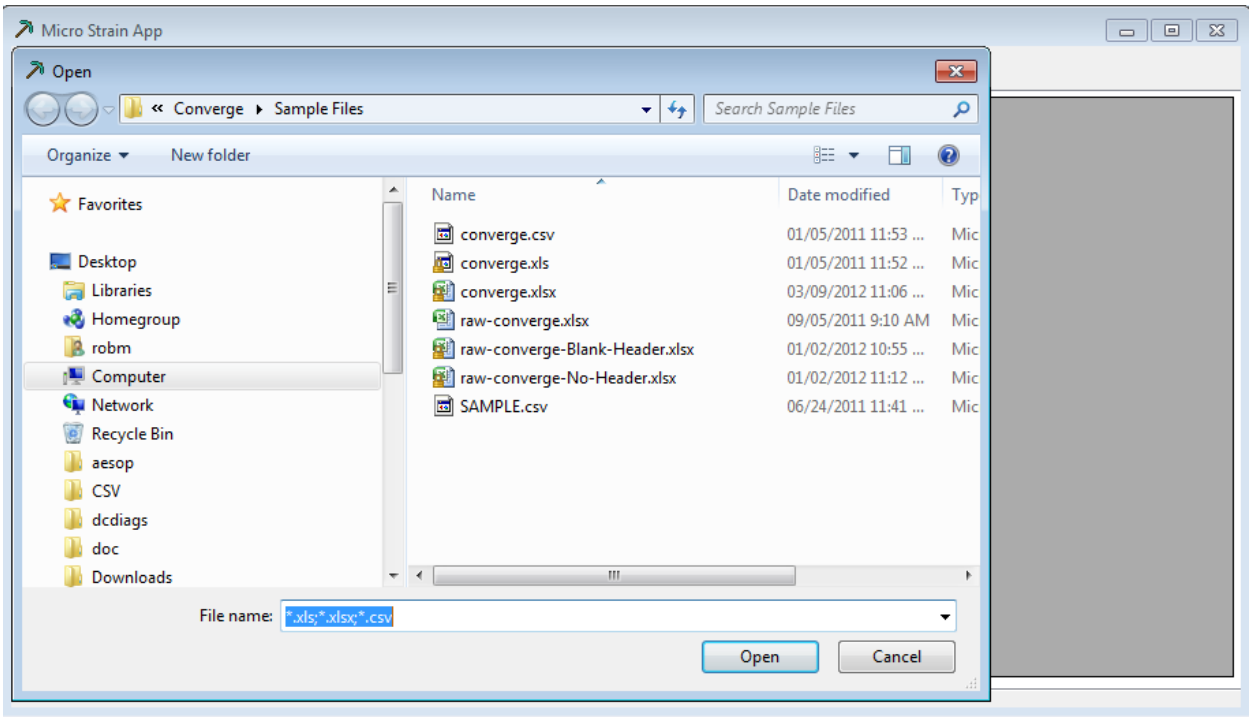
To begin using it double click on the Icon below:



This will open the progam



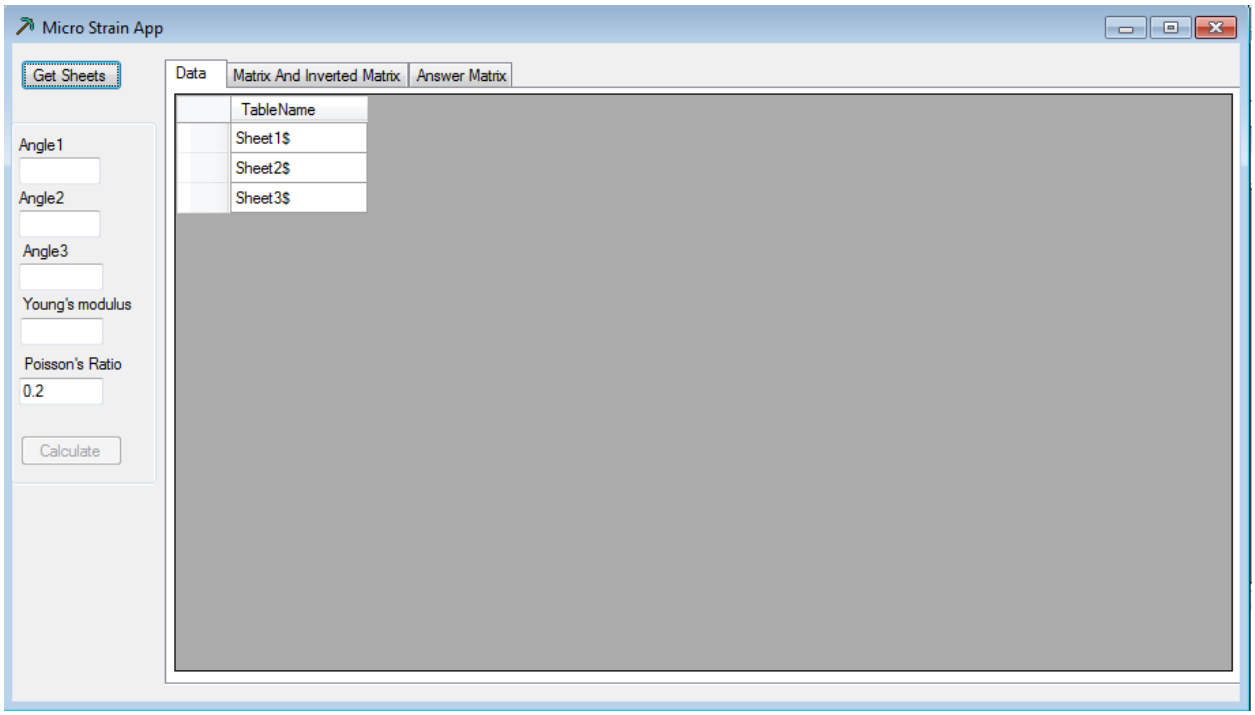
The next step is to “Get Sheets” by pressing the button in the top left corner which leads to:



Here you can see the sample files in .CSV .XLS and .XLSX format.

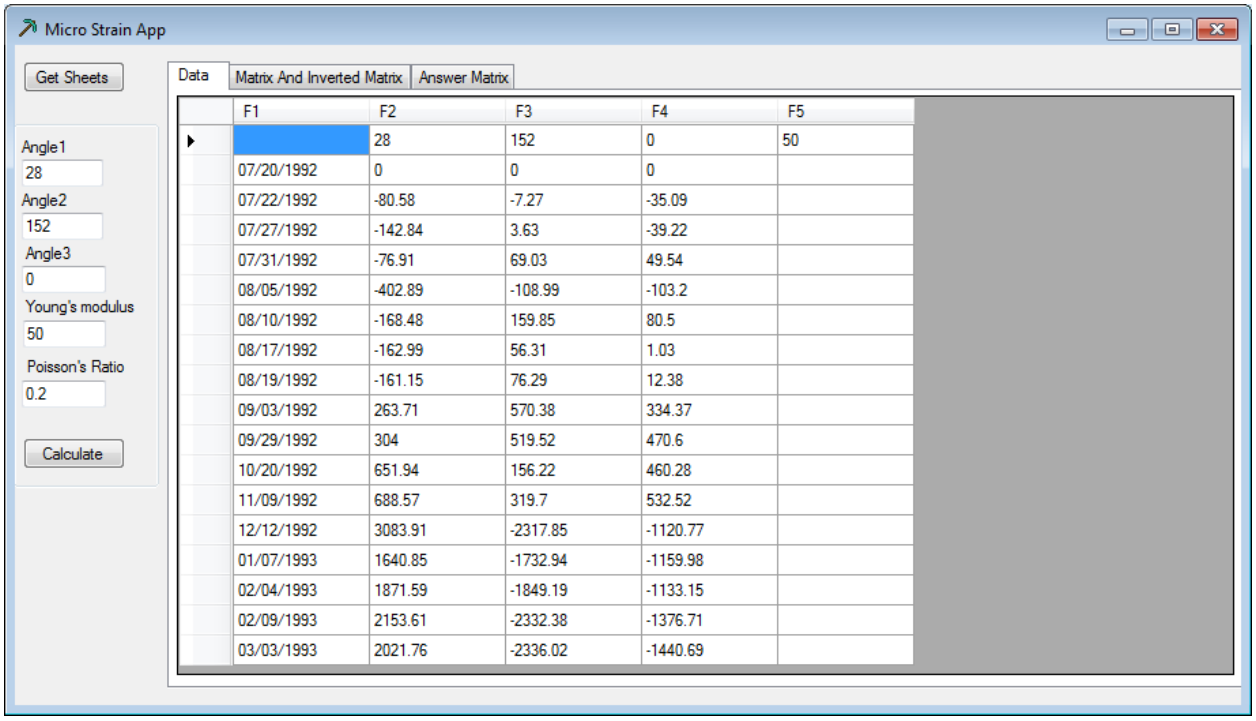
The two called raw-*.xlsx are counter examples that show bad or malformed data files.

Select a file



The Sheet1$ etc show the work sheets in the EXCEL workbook (.XLS, .XLSX) if you choose .CSV you will go straight to the spread sheet view, In the above case select the WorkSheet where your data is (typically Sheet1$ … or whatever you called your work sheet)

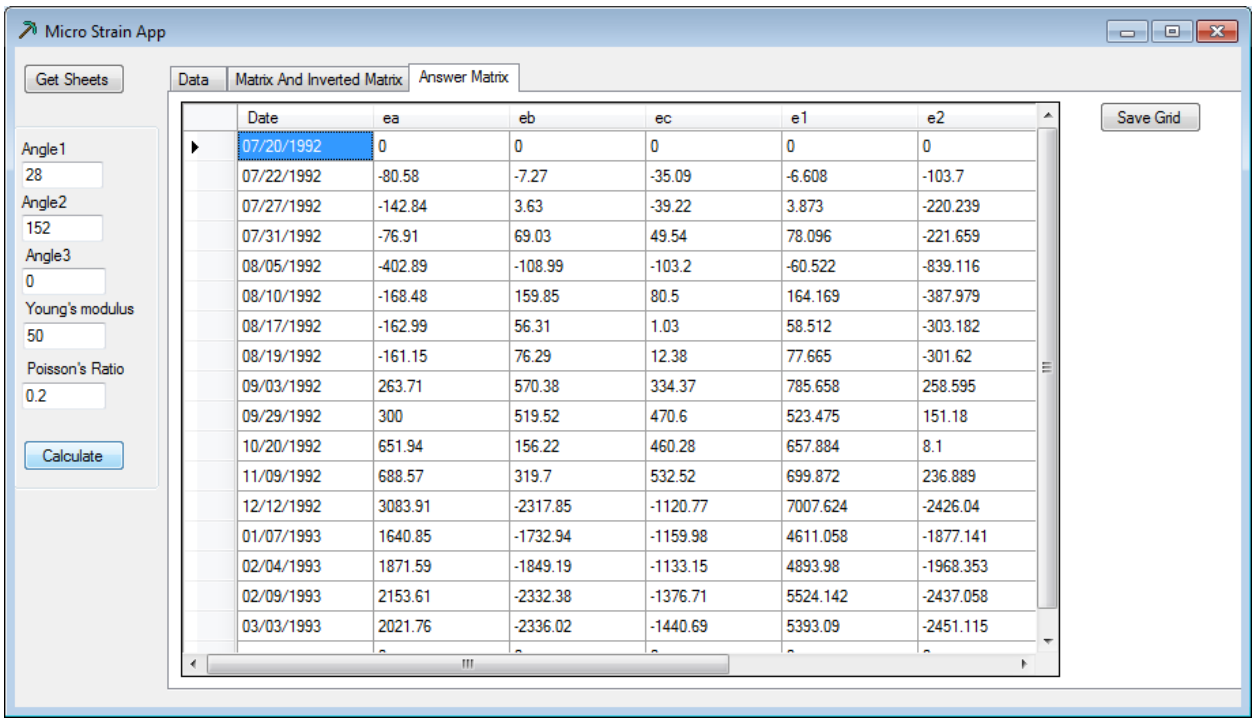
And you will get the Spreadsheet view:



F1	F2	F3	F4	F5
	28	152	0	50
07/20/1992	0	0	0	
07/22/1992	-80.58	-7.27	-35.09	
07/27/1992	-142.84	3.63	-39.22	
07/31/1992	-76.91	69.03	49.54	
08/05/1992	-402.89	-108.99	-103.2	
08/10/1992	-168.48	159.85	80.5	
08/17/1992	-162.99	56.31	1.03	
08/19/1992	-161.15	76.29	12.38	
09/03/1992	263.71	570.38	334.37	
09/29/1992	304	519.52	470.6	
10/20/1992	651.94	156.22	460.28	
11/09/1992	688.57	319.7	532.52	
12/12/1992	3083.91	-2317.85	-1120.77	
01/07/1993	1640.85	-1732.94	-1159.98	
02/04/1993	1871.59	-1849.19	-1133.15	
02/09/1993	2153.61	-2332.38	-1376.71	
03/03/1993	2021.76	-2336.02	-1440.69	

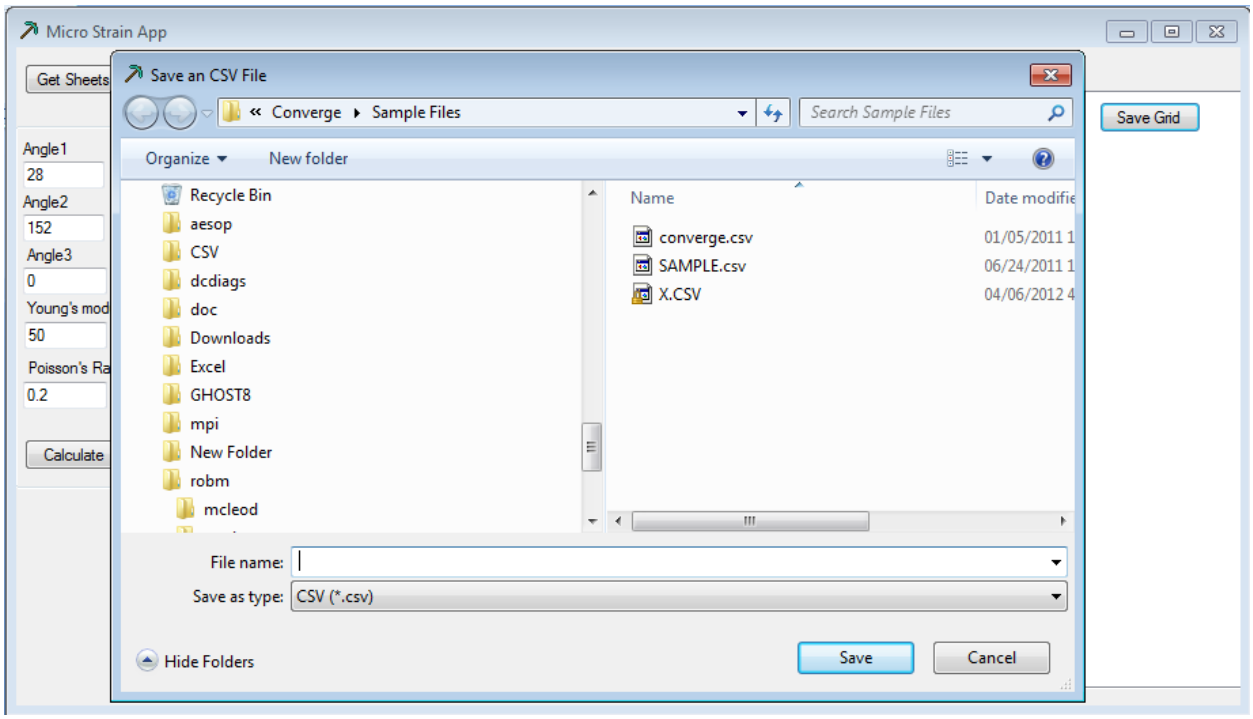
Note the Angle and Young's modulus at the top of the sheet Poisson's ratio is set at 0.2 . All of the parameters can be modified after at this point if needed.

Click calculate:

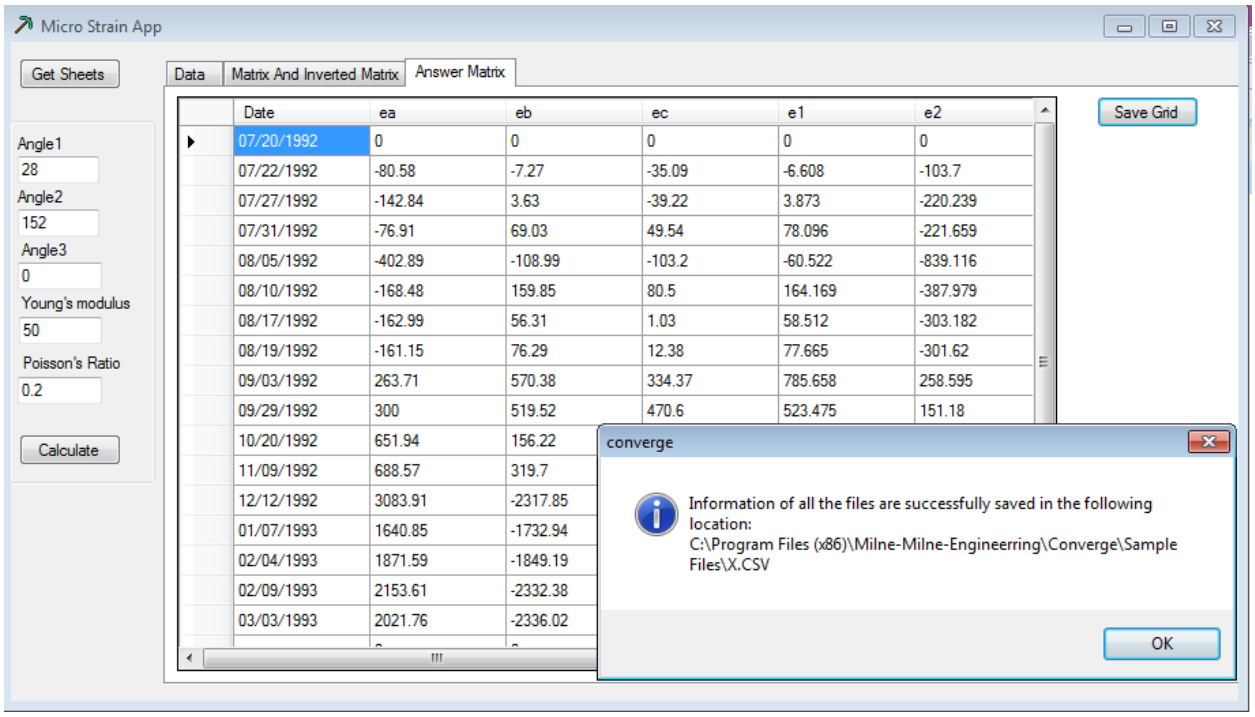


Date	ea	eb	ec	e1	e2
07/20/1992	0	0	0	0	0
07/22/1992	-80.58	-7.27	-35.09	-6.608	-103.7
07/27/1992	-142.84	3.63	-39.22	3.873	-220.239
07/31/1992	-76.91	69.03	49.54	78.096	-221.659
08/05/1992	-402.89	-108.99	-103.2	-60.522	-839.116
08/10/1992	-168.48	159.85	80.5	164.169	-387.979
08/17/1992	-162.99	56.31	1.03	58.512	-303.182
08/19/1992	-161.15	76.29	12.38	77.665	-301.62
09/03/1992	263.71	570.38	334.37	785.658	258.595
09/29/1992	300	519.52	470.6	523.475	151.18
10/20/1992	651.94	156.22	460.28	657.884	8.1
11/09/1992	688.57	319.7	532.52	699.872	236.889
12/12/1992	3083.91	-2317.85	-1120.77	7007.624	-2426.04
01/07/1993	1640.85	-1732.94	-1159.98	4611.058	-1877.141
02/04/1993	1871.59	-1849.19	-1133.15	4893.98	-1968.353
02/09/1993	2153.61	-2332.38	-1376.71	5524.142	-2437.058
03/03/1993	2021.76	-2336.02	-1440.69	5393.09	-2451.115

We have the answer matrix which can be saved via the “Save Grid” button top right.



Then save it named whatever you like.



An information widow pops up telling you where the file is saved.

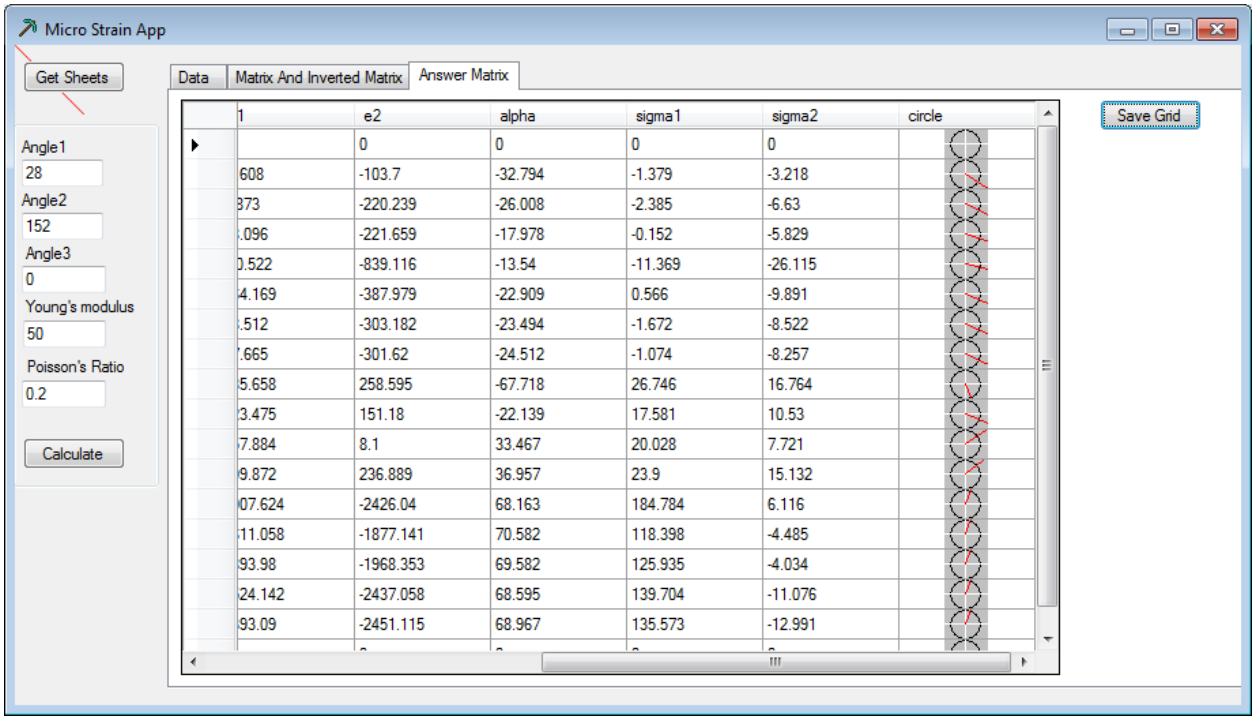
The last column on the far right (scroll over in the answer matrix) gives a graphic if the direction on maximum force change: for example



would indicate the maximum force change comes from ~-26deg

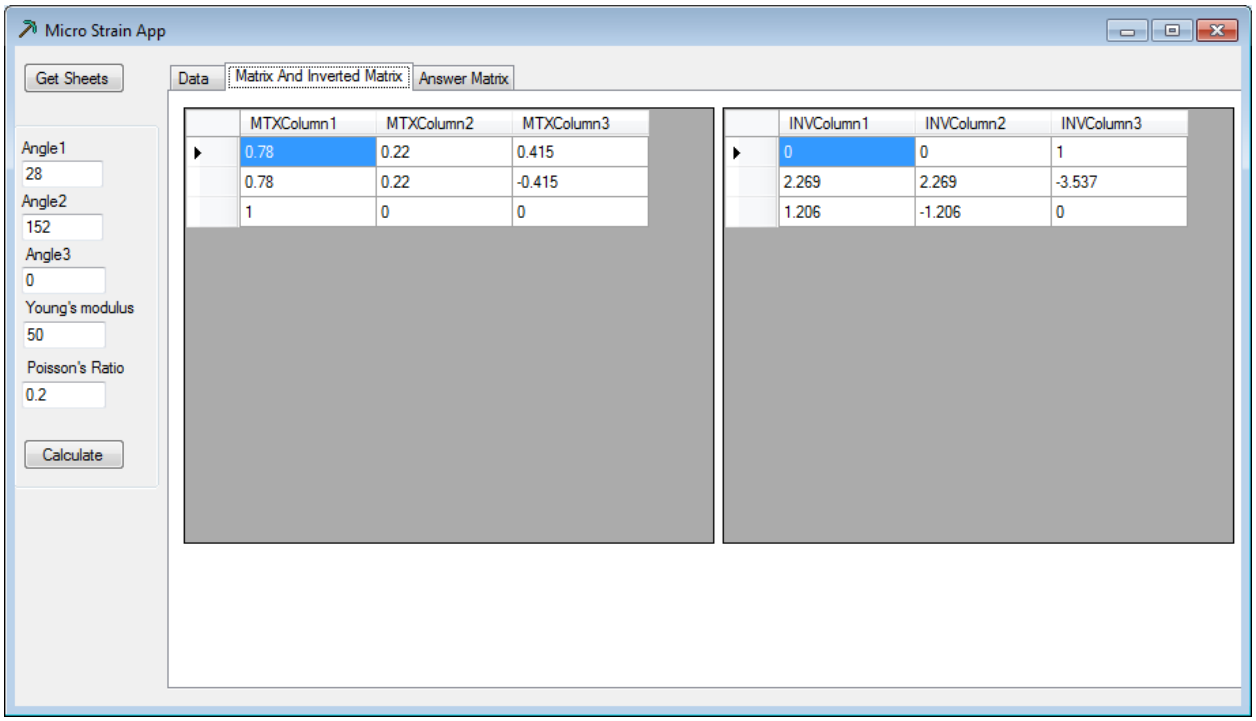


~ -160deg



1	e2	alpha	sigma1	sigma2	circle
	0	0	0	0	
608	-103.7	-32.794	-1.379	-3.218	
373	-220.239	-26.008	-2.385	-6.63	
.096	-221.659	-17.978	-0.152	-5.829	
0.522	-839.116	-13.54	-11.369	-26.115	
4.169	-387.979	-22.909	0.566	-9.891	
.512	-303.182	-23.494	-1.672	-8.522	
.665	-301.62	-24.512	-1.074	-8.257	
5.658	258.595	-67.718	26.746	16.764	
3.475	151.18	-22.139	17.581	10.53	
7.884	8.1	33.467	20.028	7.721	
9.872	236.889	36.957	23.9	15.132	
07.624	-2426.04	68.163	184.784	6.116	
11.058	-1877.141	70.582	118.398	-4.485	
93.98	-1968.353	69.582	125.935	-4.034	
24.142	-2437.058	68.595	139.704	-11.076	
93.09	-2451.115	68.967	135.573	-12.991	

As well middle tab “Matrix And Inverted Matrix” is informational values of matrices used during the calculation:



Previous Documentation:

CONVERGENCE PROGRAM - VERSION 1.0

The program 'converge.exe' has been designed to assist in the interpretation of closure station data. For input, the program takes three values of strain, a direction angle for each strain, Young's Modulus and Poisson's Ratio. A default value for Poisson's Ratio is set at .2, however, the user should realize that this value can have a significant effect on results. It has been suggested that Poisson's Ratio for a rock mass can be significantly greater than for an intact sample.

Program output consists of principal strain values and orientation, as well as principal stresses. The theory used for obtaining principal strains and orientation is based on a matrix inversion and solution taken from "Introduction to Rock Mechanics" by Goodman (1989). The resolution of principal stresses from strains applies the Kirsch solution taken from "Rock Mechancis for Underground Mining" by Brady & Brown, (1985). This solution assumes a circular drift geometry and an elastic rock mass response. The drift will not have a circular cross section, however, and this will produce some inaccuracy. When the elastic response of the rock mass appears to have been exceeded, the orientation and magnitude of principal strains may still

provide useful data.

The program is designed to take spreadsheat input with a format shown below and given in the file 'sample.prn'.

```
          28.00   152.00    0.00    75.00
20-Jul-92    0.00     0.00    0.00
22-Jul-92  -80.58    -7.27  -35.09
27-Jul-92 -142.84     3.63  -39.22
31-Jul-92  -76.91    69.03   49.54
05-Aug-92 -402.89  -108.99 -103.20
10-Aug-92 -168.48   159.85   80.50
17-Aug-92 -162.99    56.31    1.03
19-Aug-92 -161.15    76.29   12.38
03-Sep-92  263.71   570.38  334.37
29-Sep-92  304.00   519.52  470.60
20-Oct-92  651.94   156.22  460.28
09-Nov-92  688.57   319.70  532.52
12-Dec-92 3083.91 -2317.85 -1120.77
07-Jan-93 1640.85 -1732.94 -1159.98
04-Feb-93 1871.59 -1849.19 -1133.15
09-Feb-93 2153.61 -2332.38 -1376.71
```

03-Mar-93 2021.76 -2336.02 -1440.69

The first column consists of the dates at which measurments were taken. The first column adjacent to the dates are the measurments in microstrain. The value at the top of the second column, 28, is the corresponding measuring direction, in degrees, measured counter-clockwise from horizontal. 152 is the measuring direction for the second column of microstrain measurements and 0 (horizontal) is the third measuring direction for the final column of microstrain measurements. The single value in the fifth column, 75.00, is the input value of Young's Modulus, given in GPa.

This format must be followed for all input files. There must be no spaces in the date column, however, as many spaces as desired can be used between data entries. Do not skip any data entries; ie, input a date without corresponding measurements - this would throw the data reading out of sync. The program is designed to take input from exported .prn files from QPRO spreadsheet files. The program has the option of using a default value for Poisson's Ratio of .2 or any other value the user makes up.

The program displays results on the screen and the user has the option of creating a spreadsheet file '.sss' or a print file, '.ppp'. To provide complete screen and print output, the program asks for a station name, the

viewing direction for the orientations (N,S,E etc.) and the measuring length of each strain measuring direction. (The viewing direction can be put in with the file execution step).

The spreadsheet file consists of 6 columns of data. The first column repeats the reading dates, column 2 and 3 are the major and minor principal strain values and column 4 is the maximum principal strain (and stress) direction. Columns 5 and 6 are the principal stress values. No identification or background data is provided with the spreadsheet file and it is designed to be imported into a spreadsheet.

This program has been developed by Milne & Milne Engineering in conjunction with the UBC Mining-Geotechnical Group. Assistance has been provided by the Noranda Technology Centre and Energy Mines and Resources. If you find this program of use, please send 12 chocolate chip cookies (no nuts please) to Milne & Milne Engineering, 83 Leddy Cres., Saskatoon, Sk. S7H 3Y9.

C & D Milne.